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PILOT STUDY

Breast

Pilot Study: Blue Light Photobiomodulation for the Treatment of Complicated Wounds in Breast Surgery

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Background: Blue light photobiomodulation has demonstrated efficacy in accelerating the healing process of chronic ulcers and various dermatologic conditions. Yet, its application in enhancing the healing of complicated postsurgical breast wounds, particularly in reconstructive surgery, remains unexplored.

Methods: This pilot study included female patients who developed complicated wounds after reconstructive breast surgery at our institution between March 2022 and September 2023. All patients were treated with blue light therapy (EmoLED) in addition to standard wound care. We examined healing times, the number of EmoLED sessions required, and complication severity.

Results: Among 27 female patients (median age 54 y), 74.1% underwent mastectomy with implant-based reconstruction, whereas the remaining patients underwent contralateral second-stage reduction mammaplasty. Over a 19-month median follow-up, 35 complications were recorded: 18 skin necroses, 9 partial nipple–areolar complex necroses, and 8 wound dehiscences. The median number of EmoLED sessions required was 5, typically administered weekly. A higher body mass index ($P = 0.075$), smoking (0.093), Wise-pattern incision ($P = 0.007$), thin mastectomy skin flaps ($P = 0.034$), and subpectoral implant placement ($P = 0.033$) were associated with an increased number of EmoLED treatments required and longer healing times. The development of Clavien-Dindo grade III complications was associated with diabetes mellitus ($P = 0.001$), body mass index greater than 30 kg/m^2 ($P = 0.029$), smoking habit ($P = 0.037$), and thinner flap thickness ($P = 0.052$). Notably, no implant loss was reported.

Conclusions: The use of blue light therapy accelerated and optimized the healing process in complicated postsurgical breast wounds. Larger prospective studies are warranted to further validate these findings. *(Plast Reconstr Surg Glob Open 2025;13:e6989; doi: 10.1097/GOX.0000000000006989; Published online 28 July 2025.)*

INTRODUCTION

Photodynamic therapy has been extensively documented for its efficacy in managing chronic skin lesions,[1,2] which significantly impair patients' quality of life and result in considerable healthcare costs.[3] Among recent advancements in wound care, photobiomodulation has emerged as a promising approach.

This therapy uses nonionizing light sources, including lasers, light-emitting diodes, and broadband light within the visible spectrum, providing a noninvasive method to stimulate tissue healing.[3,4] Specifically, the red (633 nm) and blue (415 nm) light wavelengths have garnered attention for their broad range of potential therapeutic applications.[4] Blue light photobiomodulation (BLP), in particular, facilitates wound healing through a

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Limitations regarding long-term follow-up inherently exist in this article type.

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multifaceted mechanism of action. It mitigates the inflammatory process by increasing reactive oxygen species (ROS) production, promotes the proliferative phase by stimulating the transition of macrophages from the pro-inflammatory M1 phenotype to the reparative M2 phenotype, and enhances vasodilation and angiogenesis through nitric oxide metabolism and fibroblast activation.[4] These cellular and molecular processes collectively contribute to the formation of granulation tissue and reepithelialization, while concurrently and effectively reducing the bacterial burden.[5] These mechanisms contribute to accelerated wound healing.[6] BLP has been successfully applied in the management of various chronic wounds, including diabetic foot ulcers, venous leg ulcers, acne, burns, keloids, and psoriasis.[7–9] This study explored a novel application of BLP through the use of the EmoLED device for the treatment of complicated wounds following reconstructive breast surgery, aiming to evaluate the safety and efficacy of this innovative treatment in such a delicate and challenging clinical setting.

Takeaways

Question: Does blue light photobiomodulation (BLP), in combination with standard wound care, enhance healing outcomes in complicated postoperative breast reconstructive surgery wounds?

Findings: In this pilot study, BLP with the EmoLED device proved safe, painless, and versatile for treating complicated postbreast reconstructive surgery wounds. With a median healing time of 5 weeks, it prevented implant loss, reduced inflammation and wound exudation, and promoted granulation tissue development and high-quality scar tissue formation, even in high-risk cases.

Meaning: BLP seems to be a promising therapeutic option for managing complicated breast surgical wounds, warranting further research to validate its benefits in larger patient cohorts.

MATERIALS AND METHODS

Study Design and Device

The study used the EmoLED device, which emits blue light within the 410- to 430-nm wavelength range (Fig. 1). This portable light-emitting diode device, powered by rechargeable batteries, delivers a targeted power density of 120 mW/cm² over a circular area with a diameter of 5 cm (19.6 cm²) for a duration of 120 seconds, ensuring systematic and uniform coverage of the entire lesion. Wounds larger than 20 cm² require multiple sequential applications to ensure full coverage. Noncontact illumination of the wound bed is achieved from a distance of approximately 4 cm, delivering an energy density dose of 7.2 J/cm². BLP was administered following standard wound cleaning and debridement procedures, with treatment durations ranging from 4 to 8 weeks, tailored to individual patient responses and wound healing progression.

Patients Selection

This pilot study included female patients who developed complicated surgical wounds following breast reconstructive procedures and demonstrated limited response to initial conventional treatments. All the enrolled patients were consecutively treated with the EmoLED device alongside standard complicated wound care protocols at the Veneto Institute of Oncology between March 2022 and September 2023. The study adhered to the principles outlined in the Declaration of Helsinki and the Good Clinical Practice guidelines. Written informed consent was obtained from all participants.

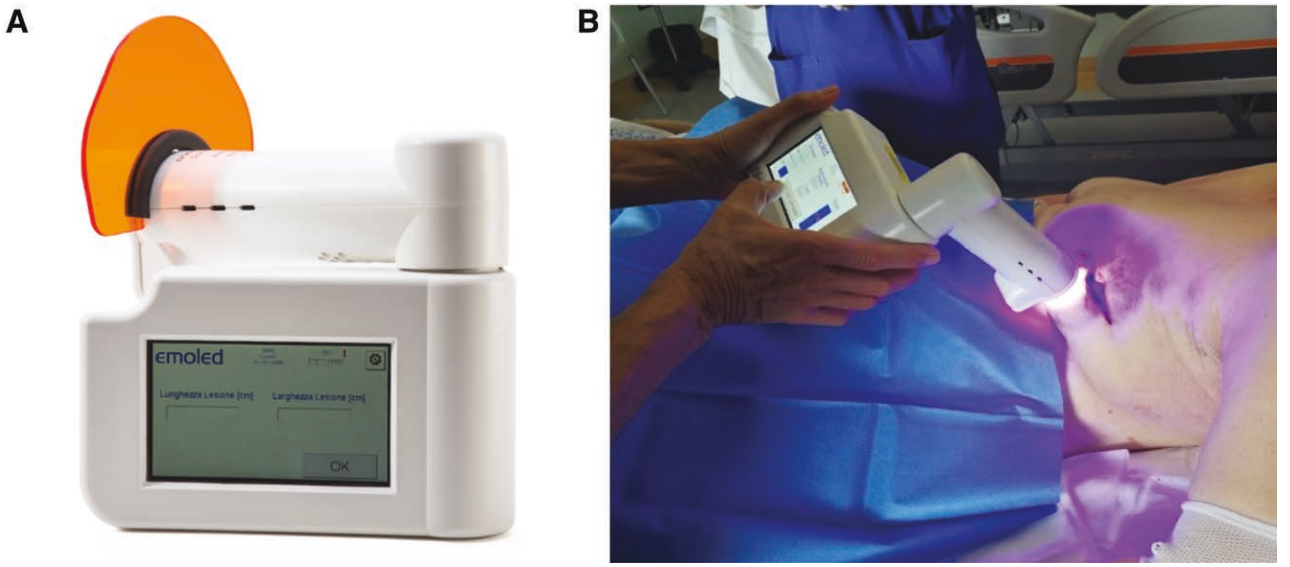


Fig. 1. The EmoLED device and an example of its clinical application. A, The EmoLED device (EmoLED SRL, Sesto Fiorentino, Florence, Italy). B, An example of blue light treatment delivered by the EmoLED device.

Data Collection

The surgical procedures included postmastectomy implant reconstruction following oncological or prophylactic surgery and contralateral symmetry procedures after breast cancer (BC) surgery. All patients underwent a thorough preoperative plastic surgery evaluation, including clinical and radiological assessment of breast flap thickness. We used the breast tissue coverage Rancati classification identifying type 1 (up to 1 cm), type 2 (1–2 cm), and type 3 (more than 2 cm).[10] All patients were subsequently monitored in an outpatient setting until complete wound healing. Additional follow-up evaluations were conducted at 6 weeks and 6 months after outpatient clinic discharge to assess long-term outcomes.

Objective scar rating was conducted at least 1 year after surgery using the Vancouver Scar Scale[11] (VSS). Two observers (M.F. and M.T.) jointly performed the evaluations, resolving any discrepancies through consensus.

Additionally, a dedicated survey was administered to both patients and surgeons to assess the overall discomfort associated with EmoLED treatment (ET). Responses were quantified using a Likert scale, ranging from 1 (no discomfort) to 4 (highest discomfort).

Surgical complications were classified according to the Clavien-Dindo classification.[12] The localization of wound complications (skin, nipple–areolar complex [NAC], or suture line), the duration and modalities of ETs, and any associated conventional therapeutic approaches were systematically documented. Wound surface areas were systematically measured at their maximum extent before healing and categorized as less than 20 cm^2 or 20 cm^2 and greater. Healing time was defined as the period between the date of surgery and the achievement of complete wound healing and full reepithelialization. A minimum follow-up period of 12 months was guaranteed.

Statistical Analyses

Continuous variables were presented as medians with interquartile ranges (IQR) or ranges, as appropriate. Categorical variables were expressed as frequencies and percentages (%). Correlations between continuous variables were investigated using the Pearson correlation coefficient. Comparisons of categorical variables were conducted using the chi-square test or the Fisher exact test. The Wilcoxon rank-sum test or Kruskal–Wallis test were used for 2-group or multiple-group comparisons, respectively. Linear regression analyses were performed to evaluate the association between potential risk factors and complications. Statistical significance was set at a *P* value of less than 0.05. All statistical analyses were performed using the R software.[13]

RESULTS

Patient Characteristics

This case series included 27 female patients with a median age of 54 years (range 37–66 y) (Table 1). The median body mass index (BMI) was 27 kg/m^2 (IQR: 25–28.5 kg/m^2). Five patients presented significant comorbidities, specifically 3 cases of diabetes mellitus (DM) and 2 cases of rheumatoid arthritis. Additionally, 8 patients were active smokers, and 4 had received neoadjuvant chemotherapy. Three patients were carriers of pathogenic mutations: 1 with BRCA1 mutation, and 2 with a BRCA2 mutation.

Table 1. Patient Characteristics and Surgical Details

Patients' Characteristics, n/N (%)	Overall (N = 27)
Age, y	
Median (IQR)	54 (44–60)
Range	37–66
BMI	
Median (IQR), kg/m^2	27 (25–28.5)
<18	0/27 (0.0)
18–25	6/27 (22.2)
25–30	16/22 (59.3)
>30	5/22 (18.5)
BRCA1–2 gene mutation	
No	24/27 (88.9)
Yes	3/27 (11.1)
Significant comorbidities	
DM	3/27 (11.1)
Rheumatoid arthritis	2/27 (7.4)
Smoking habit	
No	19/27 (70.4)
Yes	8/27 (29.6)
Neoadjuvant chemotherapy	
No	23/27 (85.2)
Yes	4/27 (14.8)
Procedures	
Unilateral	23/27 (85.2)
Bilateral	4/27 (14.8)
Surgical details, n/N (%)	Overall (N = 31)
Type of intervention	
Nipple-sparing skin-reducing mastectomy	15/31 (48.4)
Second-stage contralateral balancing reduction mammoplasty	7/31 (22.6)
Skin-reducing mastectomy	6/31 (19.4)
Nipple-sparing skin-sparing mastectomy	3/31 (9.7)
Type of surgical incision	
Wise pattern	28/31 (90.3)
Lateral radial	3/31 (9.7)
Implant	
Yes	24/31 (77.4)
No	7/31 (22.6)
Type of implant	
Polyurethane implant	17/24 (70.8)
Silicon implant	4/24 (16.7)
Tissue expander	3/24 (12.5)
Implant site	
Subpectoral	19/24 (79.2)
Prepectoral	5/24 (20.8)
Flap thickness	
Rancati type I	8/24 (33.3)
Rancati type II	11/24 (45.8)
Rancati type III	5/24 (20.8)

Surgical Procedures

Overall, 20 (74.1%) patients underwent mastectomy with associated implant-based reconstruction. Of these, 19 underwent BC treatment, whereas 1 patient had a prophylactic mastectomy due to a BRCA2 gene mutation.

Seven patients underwent contralateral balancing reduction mammaplasty as a second-stage reconstructive surgical procedure following primary mastectomy for BC. Four procedures were bilateral, resulting in a total of 31 surgically treated breasts.

The surgical interventions included 15 nipple-sparing and skin-reducing mastectomies, 7 reduction mammaplasties, 6 skin-reducing mastectomies, and 3 nipple-sparing mastectomies. In 7 cases involving large ptotic breasts, a free NAC graft was performed. The most common surgical approach was the Wise pattern incision (used in 28 cases, 90.3%), with lateral radial access used in only 3 cases.

Implant-based reconstructions were performed in 24 breasts. Implants were placed under the pectoralis major muscle using the dual-plane technique in 19 cases, including 12 polyurethane implants, 4 silicone implants, and 3 tissue expanders. Prepectoral implant placement was performed in 5 cases, and in all of them, a polyurethane implant was chosen. No synthetic or biological acellular dermal matrix was used to cover the implants. The mastectomy flap thickness was classified as type 1 in 8 of 24 cases, type 2 in 11 of 24 cases, and type 3 in 5 of 24 cases, with a median thickness of 14.5 mm (IQR: 7.7–19.2 mm). Subpectoral reconstruction was predominantly performed in patients with thinner flaps (types 1 and 2) and was the preferred approach for patients with a strong smoking history. Prepectoral reconstruction was exclusively performed in patients with no smoking history and with adequate flap thickness (type 3), in accordance with local protocols. In the only 3 cases where a tissue expander was placed in the subpectoral plane, all patients were classified as obese (BMI >30 kg/m^2), had a strong smoking history, and presented with very thin skin flaps (type 1).

Table 2. Complication Type, Grade, and Management

Complications,* n/N (%)	
Type of complications	
Skin flap necrosis	18/35 (51.4)
NAC necrosis	9/35 (25.7)
Suture dehiscence	8/35 (22.6)
Wound area, cm^3	
Median (IQR)	19 (17–24)
<20 cm^3	15/27 (55.5)
≥ 20 cm^3	12/27 (44.4)
Grade of complications	
Clavien-Dindo I	17/27 (63.0)
Clavien-Dindo II	5/27 (18.5)
Clavien-Dindo III	5/27 (18.5)
Clavien-Dindo IV	0/27 (0.0)
Complications management, n/N (%)	
ET sessions schedule	
Once a week	24/27 (88.9)
Twice a week	3/27 (11.1)
Intraoperative	3/27 (11.1)
ET sessions number	
2	4/27 (14.8)
3	4/27 (14.8)
4	6/27 (22.2)
5	9/27 (33.3)
≥6	4/27 (14.8)
Associated treatments	
Advanced wound care and dressing	27/27 (100)
Escharectomy	20/27 (74.1)
Surgical wound revision	5/27 (18.5)
Antibiotic treatment	5/27 (18.5)
NPWT	5/27 (18.5)
HBOT	1/27 (3.7)

*A total of 35 complications were observed in 27 breasts. All complications were unilateral.

Postsurgical Complications

Over a median follow-up of 19 months (range 12–30 mo), a total of 35 complications were reported, all unilateral, among the 27 analyzed patients (31 treated breasts) (Table 2). Skin necrosis was observed in 18 patients, predominantly affecting the medial skin flaps following Wise pattern skin-reducing mastectomy incisions; 14 of the 18 cases occurred in this area. Partial NAC necrosis was reported in 9 patients, including 1 case of subtotal NAC necrosis. Wound dehiscence was observed in 8 patients, most commonly at the inverted T-junction of Wise pattern incisions, accounting for 7 of the 8 cases. Notably, all ischemic-necrotic complications occurred in patients who underwent mastectomy with implant-based reconstruction, except for 2 cases observed in patients who underwent a second-stage balancing reduction mammaplasty. In contrast, the latter group was primarily affected by surgical suture dehiscence, which accounted for 5 of the 7 cases ($P < 0.001$).

Complication Management

The median number of ET sessions was 5 (IQR: 3–5), typically delivered at 1 session per week, resulting in a median treatment duration of 5 weeks. An exception was observed in a case of very severe complications (pyoderma gangrenosum, described in detail in the Notable Case C section), which required 2 sessions per week. In 3 cases of second-stage reduction mammaplasty with associated NAC free grafts, the first BLP was also applied intraoperatively to prevent ischemic complications of the NAC, which were successfully avoided. All ETs following implant-based reconstructions were performed with the implant or tissue expander in place. In the majority of cases, ETs were initiated between 7 and 14 days after surgery (range 5–21 d, excluding intraoperative use). The duration of each ET session ranged from 120 to 360 seconds (mean 178 s), depending on the wound surface area. With a median wound area of 19 cm^2 (IQR: 17–24 cm^2), 15 wounds measured less than 20 cm^2, requiring a 120-second treatment, whereas 12 patients had wounds 20 cm^2 and greater, necessitating longer ET sessions. A higher BMI showed a trend toward significance in requiring more ETs ($P = 0.075$). The incision type significantly influenced the number of required sessions, with Wise pattern incisions necessitating more ETs to achieve complete healing ($P = 0.007$). Cases with subpectoral implant placement also required a significantly higher number of BLP sessions to obtain full reepithelialization ($P = 0.033$). Notably, smoking history slightly impacted the number of ETs required

($P = 0.093$). Among patients undergoing implant-based reconstruction, a significantly greater number of ETs was performed in those with reduced skin flap thickness, with the most remarkable difference noted between type 1 and type 3 flaps (type 1: mean 5.71 sessions; type 2: 4.25 sessions; type 3: 3 sessions; $P = 0.034$).

Conventional standard wound treatments, including Vaseline gauze application, gentamicin ointment, and silver sulfadiazine, were used in all cases as complementary to ETs. Negative-pressure wound therapy (NPWT) was applied in 5 cases without benefit before ETs. One patient with severe complications also received hyperbaric oxygen therapy (HBOT) (see Notable Case C). Seventeen patients required only advanced wound care and dressing in the outpatient setting, without pharmacological treatment or surgical/radiological intervention (Clavien-Dindo grade I). In 5 cases of superficial wound infection, conservative antibiotic treatment was successfully administered (grade II). Five patients underwent wound revisions (2 under local anesthesia, grade IIIa, and 3 under general anesthesia, grade IIIb, including 1 requiring a free skin graft).

The development of grade III complications was strongly associated with DM ($P = 0.001$), BMI greater than 30 kg/m^2 ($P = 0.029$), and a strong smoking history ($P = 0.037$), and was slightly associated with thinner flap thickness ($P = 0.052$). Only 1 long inpatient readmission (6 wk) was required (see Notable Case C). No cases of implant loss occurred in this series, even among high-risk patients, and complete healing was achieved without any negative effect or damage to the implant related to the use of the device. Objective scar assessment was performed at least 1 year after surgery in 23 patients (85.2% of the cohort), revealing a median VSS of 5 (IQR: 3–7).

The specific survey evaluating patient discomfort related to ET showed that all respondents (85.2% of the population) reported no discomfort, scoring 1 out of 4 on the Likert-like scale. Similarly, both primary authors (M.F. and M.T.) reported no discomfort or issues related to ET administration. No technical failures or device malfunctions were registered throughout the treatment course, and no adverse events were reported, outlining a favorable safety profile.

Healing Times

The median healing time for these complicated wounds was 5 weeks (range 2–10 wk). In all the cases, initiation of ET was followed by progressive wound healing, with complete reepithelialization observed within the follow-up period. The linear regression analysis between BMI and healing time showed a weak negative relationship ($P = 0.068$; $R^2 = 0.127$). Additionally, the average healing time was slightly higher for patients with a history of smoking compared with those who had no history of tobacco use (5.4 versus 3.9 wk).

The Wise pattern incision was significantly associated with prolonged healing times ($P = 0.018$). Subpectoral implant placement also correlated with significantly longer healing times ($P = 0.043$), as did thinner skin flaps ($P = 0.041$).

Notable Cases

Notable Case A

A 40-year-old woman with a BRCA2 gene mutation underwent bilateral prophylactic nipple-sparing skin-reducing mastectomy, followed by dual-plane breast reconstruction using 315 mL anatomical silicone implants. She developed early postoperative necrosis of the inner mastectomy skin flap and the NAC of the left breast. Initial management included standard care with Vaseline gauzes, gentamicin ointment, and silver sulfadiazine. Additionally, NPWT was applied for 2 weeks; however, no improvement was observed. Consequently, the patient was treated with BLP using the EmoLED device. This approach resulted in significant wound bed improvement, with a reduction in lesion depth and revitalization of the perilesional skin. Complete reepithelialization of the inner quadrant of the left breast was achieved after 1 month of weekly ETs. (**See figure, Supplemental Digital Content 1**, which displays ischemia and necrosis of the inner left mastectomy skin flap and the NAC, observed 1 month after surgery and following 2 weeks of NPWT [A], initial improvements in the wound bed, noted 1 week after the first application of BLP using the EmoLED device [B], onset of the reepithelialization process after 4 sessions of BP [C], advanced reepithelialization observed 1 week after the final session of BLP [D], and complete reepithelialization achieved 3 months after surgery and 1 month after the conclusion of BLP [E], **https://links.lww.com/PRSGO/E211**.)

Notable Case B

A 47-year-old woman with a BRCA2 gene mutation and right breast triple-negative invasive BC underwent neoadjuvant chemotherapy, followed by bilateral nipple-sparing skin-reducing mastectomy (prophylactic for the left breast). Dual-plane breast reconstruction was performed using 475-mL anatomical silicone implants. She developed early postoperative necrosis of the inner mastectomy skin flap and the NAC of the right breast. Initial management included standard wound care and NPWT; however, no improvement was observed. The local treatment was then shifted to silver sulfadiazine until eschar detachment. Subsequently, the necrotic area was treated with 7 weeks of BLP using the EmoLED device, resulting in more than 90% wound size reduction and near-complete reepithelialization, with high-quality scar formation. Complete reepithelialization was achieved 3 months after surgery. (**See figure, Supplemental Digital Content 2**, which displays early postoperative ischemia of the medial mastectomy skin flap and NAC of the right breast [A]; demarcated necrosis of the ischemic tissue [B]; initial eschar detachment following NPWT and silver sulfadiazine treatment [C]; application of BLP on the wound bed, with a total of 5 treatment sessions [D]; gradual formation of granulation tissue and high-quality scar tissue, with the patient reporting partial recovery of cutaneous sensitivity in the treated area [E]; and complete reepithelialization observed 3 months after surgery [F], **https://links.lww.com/PRSGO/E212**.)

Notable Case C

A 45-year-old woman with a medical history of ulcerative rectocolitis and Hodgkin lymphoma developed pyoderma gangrenosum at her port-a-cath site 6 years earlier, during neoadjuvant chemotherapy for triple-negative right BC. This condition progressed to mediastinitis, requiring extensive management. After clinical resolution, she underwent a right mastectomy without reconstruction. Five years postmastectomy, the patient pursued breast reconstruction with lipofilling of the right breast and contralateral reduction mammaplasty. Two weeks postreconstruction, she developed hyperpyrexia and elevated inflammatory markers, along with severe skin changes in the left breast. A wound biopsy confirmed a diagnosis of pyoderma gangrenosum. Treatment included a regimen of methylprednisolone,[14] colchicine, and 30 sessions of HBOT. This was supplemented with 12 sessions of ET, administered twice weekly during the hospital stay. This resulted in progressive wound improvement and complete reepithelialization, achieved 4 months after surgery. (**See figure, Supplemental Digital Content 3**, which displays pyoderma gangrenosum on the left breast, observed 2 weeks after reduction mammoplasty [A]; left breast 1 week after the beginning of HBOT and BLP [B]; improvement in the left breast 3 weeks after starting treatment with HBOT (15 sessions), BLP (6 treatments), and advanced dressings [C]; significant healing of the left breast 6 weeks after treatment with HBOT (30 sessions), BLP (12 treatments), and advanced dressing [D]; and complete reepithelialization observed 4 months after surgery [E], **https://links.lww.com/PRSGO/E213**.)

DISCUSSION

This case series highlighted the potential efficacy of BLP in managing complicated wounds following reconstructive breast surgery. The EmoLED device demonstrated significant clinical utility, being compact, safe, painless, and user-friendly[15] in both outpatient and operating room settings. Its application promoted granulation tissue formation and accelerated wound reepithelialization across all cases described, aligning with prior studies.[16,17] Although NPWT is widely recognized for its efficacy in managing acute and chronic wounds,[18,19] the innovative addition of BLP has shown synergistic effects, enhancing wound healing outcomes, especially in challenging clinical contexts.[20]

BLP Mechanisms

BLP promotes wound healing through multiple mechanisms. By absorbing flavins, it induces an increase in ROS production, thereby modulating the inflammatory response, reducing its duration, and accelerating its resolution. Additionally, BLP facilitates the proliferative phase by stimulating macrophage activation, while also promoting vasodilation and angiogenesis through nitric oxide metabolism and fibroblast activation.[4] BLP enhances cellular respiration and adenosine triphosphate production,[21] providing the energy necessary for the proliferative and remodeling stages, thereby supporting granulation tissue formation and wound healing. Beyond its regenerative effects, BLP exhibits antimicrobial properties by reducing bacterial load and inhibiting biofilm formation through photoexcitation of intracellular porphyrins, leading to the ROS-mediated destruction of microbial cells.[5] Notably, this antimicrobial effect is achieved with fewer detrimental effects on healthy tissues compared with ultraviolet irradiation.[5] BLP is believed to also have analgesic properties, potentially by modulating nociceptive pathways and reducing inflammatory mediators associated with pain. Some reports indicate that BLP may induce the release of endogenous opioids, contributing to pain relief.[22]

Risk Factors and Outcomes

The risk factors influencing postoperative complication rates following postmastectomy breast reconstruction have been extensively investigated in multicentric studies.[23–26] Among them, obesity,[27,28] smoking,[29] DM,[30] thin mastectomy skin flaps, and receiving neoadjuvant chemotherapy are well-documented contributors to postoperative complications, such as wound dehiscence, skin necrosis, and implant failure.[31] This pilot study highlighted the impact of these risk factors, which are commonly encountered among patients with severe and hard-to-treat complications. Elevated BMI, strong smoking habit, thin mastectomy skin flaps, subpectoral implant positioning, and Wise pattern incisions were associated with delayed healing and increased ET sessions required. Despite the presence of significant risk factors in our patient cohort, all cases achieved complete reepithelialization, and no implant losses were observed. Additionally, the literature establishes a clear association between DM and delayed wound healing in patients undergoing implant-based breast reconstruction,[30] with an implant failure rate of 15.2% reported in cohorts of patients with diabetes. In contrast, our study demonstrated the efficacy of BLP in patients with diabetes, achieving a 0% implant failure rate among the limited number of cases with diabetes included in this series.

Interestingly, our case series also included several active tobacco users, a category at high risk for delayed wound healing and infection after breast reconstructive surgery.[29] Despite this elevated risk, these patients achieved favorable outcomes, suggesting that BLP may offset some of the detrimental effects of smoking on wound healing.

Novelty and Clinical Implications

This study represented the first documented and promising application of EmoLED in breast surgery wound management, addressing a gap in clinical protocols for BLP. The device's ability to reduce inflammation, control bacterial burden, and promote reepithelialization aligns with previous findings in dermatologic and chronic wound care.[32,33] Our results suggest that BLP could also be expanded to intraoperative applications to prevent ischemic complications.

Limitations

The main limitations of this study stem from its retrospective, single-institution design and small sample size, which limit generalizability. The absence of a comparison group

precludes definitive conclusions about the relative efficacy of BLP compared with other modalities. Additionally, certain aspects of complicated wound management, such as associated pain and formal quality-of-life assessment, were not specifically analyzed. The choice of using the VSS may be controversial, as it was originally designed for burn scar assessment[11] and could be subject to interobserver variability.[34] Nevertheless, given the lack of standardized methods for evaluating surgical scars following BC surgery and the recognition of VSS as a reliable and valid tool for objective breast scar assessment,[34,35] its use may be justified.

CONCLUSIONS

BLP has demonstrated significant potential in enhancing wound healing processes among patients undergoing breast reconstructive surgery. BLP proved to be safe, well tolerated, and an effective adjunct to standard wound care protocols. Its ease of use, quick application, and absence of adverse effects make it a compelling option for widespread clinical adoption. Notably, even in high-risk patients with elevated BMI or smoking habits, complete healing was achieved within a reasonable timeframe. Based on these findings, we hypothesize that BLP could also be extended to advanced oncoplastic procedures and aesthetic breast surgery. Moreover, this treatment is feasible and could be useful in other clinical settings, such as prophylactic intraoperative use to prevent ischemic complications in cases of complex mastectomies. Future prospective, multicentric randomized studies are needed to confirm these findings and explore broader applications.

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DISCLOSURE

The authors have no financial interest to declare in relation to the content of this article.

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